

1. Administrative & Study Identification

Full Study Title: An Investigational Immuno-therapy Trial of Nivolumab, or Nivolumab Plus Ipilimumab, or Nivolumab Plus Platinum-doublet Chemotherapy, Compared to Platinum Doublet Chemotherapy in Patients With Stage IV Non-Small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** CheckMate 227 **Protocol Number:** CA209227 **NCT Number:** NCT02477826 **Posting ID:** 1011747896194183411 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab (Opdivo), Ipilimumab (Yervoy), and Platinum-doublet Chemotherapy (including Cisplatin) **Phase of Development:** Phase 3 (Phase Requirement: Trial must be in PHASE3) **Trial Status:** Completed (Study has been completed) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether first-line treatment with Nivolumab, or Nivolumab plus Ipilimumab, or Nivolumab plus Platinum-Doublet Chemotherapy improves Progression-Free Survival (PFS) and/or Overall Survival (OS) compared with platinum-doublet chemotherapy in patients with advanced/metastatic non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DoR), and disease-related symptom improvement (specifically evaluating the percentage of participants with symptom deterioration).

2.2 Background & Rationale Lung cancer is a leading cause of cancer-related mortality, and patients with advanced NSCLC historically have a poor long-term survival rate with standard chemotherapy. Immunotherapies targeting the PD-1 (nivolumab) and CTLA-4 (ipilimumab) pathways stimulate the body's immune system to attack tumor cells. This study aims to demonstrate that a dual immune checkpoint inhibitor combination, with or without chemotherapy, provides long-term, durable survival benefits over standard-of-care chemotherapy in the first-line setting for metastatic NSCLC, regardless of baseline PD-L1 expression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Platinum-doublet chemotherapy) **Blinding:** Open-label **Randomization:** Randomized (1:1:1 allocation dependent on PD-L1 expression levels $\geq 1\%$ or $< 1\%$) **Status:** Completed

3.2 Planned Enrollment & Duration Targeted Enrollment: 2,747 patients **Number of Sites:** Global multicenter trial **Estimated Study Start:** 05-Aug-2015 **Estimated Primary Completion:** 25-Oct-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Subjects with histologically confirmed Stage IV or recurrent NSCLC squamous or non-squamous histology, with no prior systemic anticancer therapy. 2. Subjects must have programmed death-ligand 1 (PD-L1) immunohistochemical (IHC) testing, with results, performed by the central lab during the Screening period. 3. Eastern Cooperative Oncology Group (ECOG) Performance Status of ≤ 1 . 4. Measurable disease by CT or MRI per response evaluation criteria in solid tumors version 1.1 (RECIST 1.1) criteria.

4.2 Exclusion Criteria 1. Known EGFR mutations or ALK alterations sensitive to targeted therapy. 2. Subjects with untreated Central Nervous System (CNS) metastases are excluded. 3. Subjects with an active, known, or suspected autoimmune disease are excluded. 4. Any positive test for hepatitis B virus or hepatitis C virus or human immunodeficiency virus (HIV) indicating acute or chronic infection. *(Other protocol defined inclusion/exclusion criteria apply)*

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Nivolumab (3 mg/kg every 2 weeks) + Ipilimumab (1 mg/kg every 6 weeks). * Nivolumab monotherapy (240 mg every 2 weeks). * Nivolumab (360 mg every 3 weeks) + Platinum-doublet chemotherapy (including Cisplatin [Trade Name: Platinol, ATC Code: L01XA01]). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or for up to 2 years of immunotherapy.

5.2 Concomitant & Prohibited Therapy Permitted: Routine symptom management and palliative care. **Prohibited:** Concurrent systemic anti-cancer therapies, investigational agents, and immunosuppressive doses of systemic corticosteroids.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Imaging (CT/MRI), PD-L1 Testing, Safety Labs | | **Baseline** | Day 0 | Randomization, First Dose Administration | | **Interim** | Every 2 to 6 Weeks | Safety Labs, Treatment Administration, AE Monitoring, Tumor Imaging (every 6 weeks for

first year) | | **End of Study** | Up to 2 Years or Progression |
Final Scans, Safety Follow-Up, Transition to Survival Follow-up |

(Note: Survival follow-up continues periodically after treatment discontinuation)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Progression-Free Survival (PFS) Per BICR:**

Defined as the time between the date of randomization and the date of first documented disease progression, based on BICR assessments (per RECIST v1.1), or death due to any cause, whichever occurs first based on Kaplan-Meier estimates.

Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum during the study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm.

* **Overall Survival (OS):** OS for all randomized participants is the time between randomization date and the date of death from any cause.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) Per BICR. * Duration of Response (DoR). * Percentage of Participants With Symptom Deterioration at Week 12 Assessed Via Lung Cancer Symptom Scale (LCSS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-related AEs (including immune-mediated AEs like colitis, pneumonitis, and rash) and laboratory abnormalities prior to each dose. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) established to review safety and efficacy data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all randomized efficacy outcomes; Safety Set for all participants who received at least one dose of study medication. **Sample Size**

Justification: Appropriately powered (Targeted Enrollment: 2,747) to detect significant OS and PFS differences using a hierarchical testing strategy for specific PD-L1 and Tumor Mutational Burden (TMB) subgroups. **Handling of Missing Data:** Kaplan-Meier

methodology used for time-to-event endpoints (OS, PFS) with appropriate censoring rules; patients without post-baseline scans are considered non-responders for ORR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and centralized data monitoring to ensure protocol adherence. **Audit Trail:** eCRF locking, tracking of safety reports, and rigorous Blinded Independent Central Review (BICR) for primary imaging outcomes.

11. Study Identifiers & Governance

Primary ID: CA209227 **Registry Number:** NCT02477826 **Posting ID:** 1011747896194183411 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** CheckMate 227 **Official Regulatory Title:** An Open-Label, Randomized Phase 3 Trial of Nivolumab, or Nivolumab Plus Ipilimumab, or Nivolumab Plus Platinum Doublet Chemotherapy Versus Platinum Doublet Chemotherapy in Subjects With Chemotherapy-Naïve Stage IV or Recurrent Non-Small Cell Lung Cancer (NSCLC)

12. Clinical Framework (Oncology Specific)

Primary Condition / Disease Area: Non-Small Cell Lung Cancer (Preferred Name: Metastatic non-small cell lung cancer) **Preferred UMLS Name:** Non-Small Cell Lung Carcinoma **Semantic Type:** Neoplastic Process **Classification Codes:** MeSH Code: D016938 | SNOMED ID: 457721000124104 **Diagnosis Threshold / Definition:** Histologically/cytologically confirmed advanced disease, chemotherapy-naïve. Stage IV includes: (Any T, Any N, M1a); (Any T, Any N, M1b). M1a: Lung cancer with separate tumor nodule(s) in a contralateral lobe; tumor with pleural nodules or malignant pleural (or pericardial) effusion. M1b: Distant metastasis (AJCC 7th ed.). **Medical Methodology:** Immune Checkpoint Inhibitor Therapy vs. Cytotoxic Chemotherapy **Optimization Target:** Prolonged Overall Survival (OS) and Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: First-line systemic therapy for metastatic NSCLC **Education Protocol (HCP):** Investigator training on recognizing and managing immune-related Adverse Events (irAEs) **Education Protocol (Patient):** Comprehensive informed consent regarding the unique side-effect profile of dual immunotherapy **Quality Audit Mechanism:** Centralized RECIST 1.1 radiographic assessment (BICR) to ensure unbiased progression tracking

14. Observation & Follow-Up Schedule

Total Duration: Minimum 5-year survival follow-up (up to 73.5+ months reported) **Onsite Visit Cadence:** Dosing every 2, 3, or 6 weeks depending on assigned arm **Remote Monitoring Frequency:** Regular survival tracking calls/checks post-progression **Checklist Review Interval:** Tumor assessments strictly every 6 weeks for year 1, then every 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					